

# SEED\_2026 — run002: Scaling Up the Training Set

3,000 scenes vs. run001's 500 — does more data fix the moderate/severe confusion?

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## 1. Summary

run001 (500 scenes) reached QWK 0.40, with the model reliably separating clean from ambiguous but frequently confusing the moderate and severe middle classes — plausibly because those classes had only ~300–500 training examples each. This report covers **run002**: the same pipeline and model, run on **3,000 scenes** (6x more), to test that hypothesis directly.

**Result: QWK rose from 0.40 to 0.64**, and the improvement is concentrated exactly where predicted — the moderate and ambiguous classes, which had the least data before.

## 2. The data

3,000 scenes produced 36,106 cutouts, roughly 6x run001's 6,013:

| class     | run001 (500 scenes) | run002 (3,000 scenes) | growth |
|-----------|---------------------|-----------------------|--------|
| clean     | 5,105               | 30,509                | 6.0x   |
| moderate  | 485                 | 3,200                 | 6.6x   |
| severe    | 307                 | 1,682                 | 5.5x   |
| ambiguous | 116                 | 715                   | 6.2x   |

## 3. Training

Same architecture, loss, and optimizer settings as run001's final (stable) attempt — AdamW, gradient clipping, cosine LR decay, dihedral augmentation, inverse-frequency sampling. No hyperparameter changes; the only variable is dataset size.

Training is still visibly noisy epoch-to-epoch (expected — the weighted sampler and small rare-class counts mean each epoch sees a different draw), but the *level* it oscillates around is now clearly higher and the best epoch reaches much further: **QWK 0.64** vs. 0.40.

## 4. Final model evaluation

Held-out validation split: 7,150 cutouts from scenes never used in training (proportionally larger than run001's 1,197, since the dataset itself is larger).

- **QWK: 0.64** (substantial agreement, up from 0.40 / moderate)

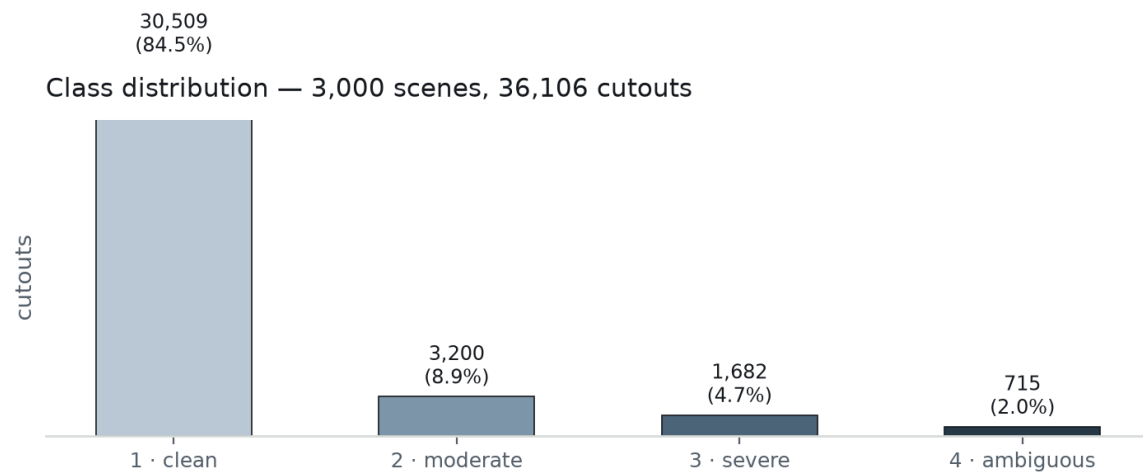


Figure 1: Class distribution

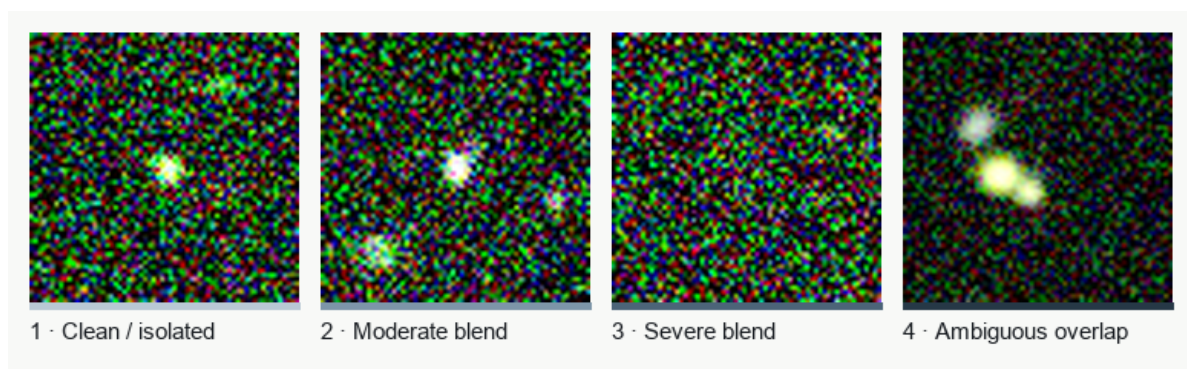


Figure 2: Example cutouts

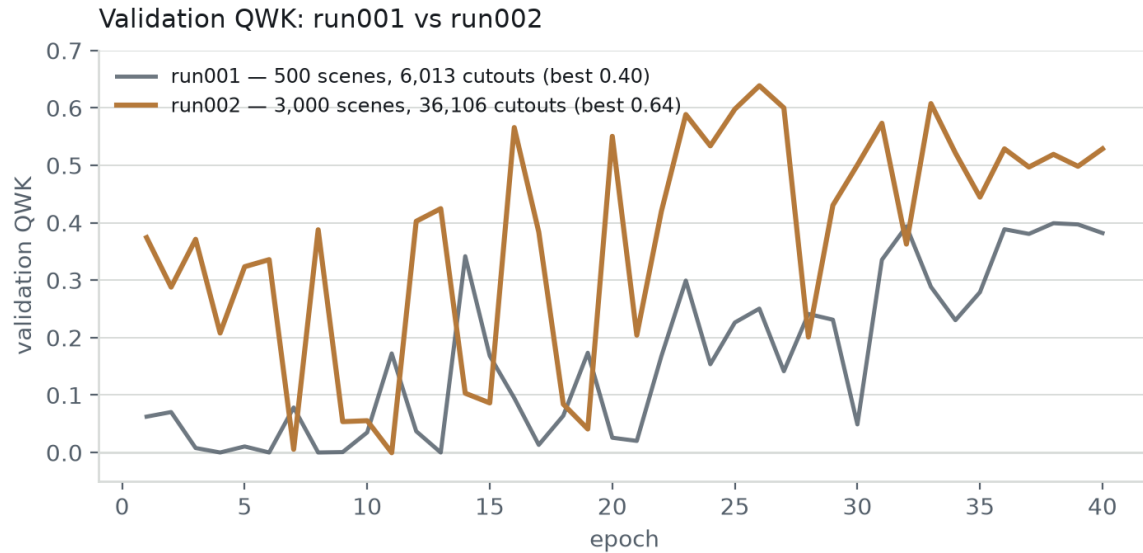


Figure 3: Validation QWK: run001 vs run002

- **Balanced accuracy: 0.509** (up from 0.373)

|              | precision | recall | f1-score | support |
|--------------|-----------|--------|----------|---------|
| clean        | 0.92      | 0.96   | 0.94     | 6019    |
| moderate     | 0.43      | 0.42   | 0.42     | 667     |
| severe       | 0.47      | 0.23   | 0.31     | 324     |
| ambiguous    | 0.79      | 0.43   | 0.56     | 140     |
| accuracy     |           |        | 0.86     | 7150    |
| macro avg    | 0.65      | 0.51   | 0.56     | 7150    |
| weighted avg | 0.85      | 0.86   | 0.85     | 7150    |

Every class improved, but not equally:

- **Ambiguous:** F1 0.25 -> 0.56 — more than doubled. This was the rarest class (116 examples) and benefited the most from 6x more data.
- **Moderate:** F1 0.19 -> 0.42 — also more than doubled.
- **Severe:** F1 0.23 -> 0.31 — improved but remains the weakest class. The confusion matrix shows severe is still most often mistaken for moderate (121 of 324) or even clean (115 of 324), suggesting this boundary is harder than data volume alone can fix — see §5.
- **Clean:** F1 0.91 -> 0.94 — already strong, modest further gain.

## 5. Interpretation and next steps

The core hypothesis from run001 is confirmed: **data volume was a real bottleneck**, not just a modeling limitation, for the moderate/ambiguous classes. Doubling down on more scenes should keep helping there.

Severe is the outlier — it improved the least relative to its 5.5x data increase, and its confusion

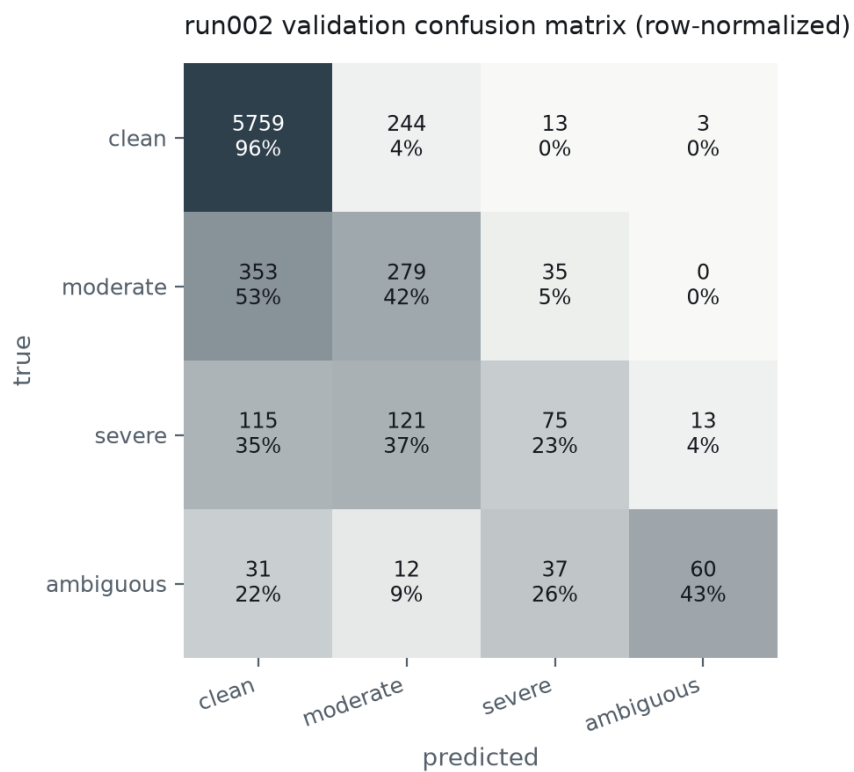


Figure 4: run002 confusion matrix

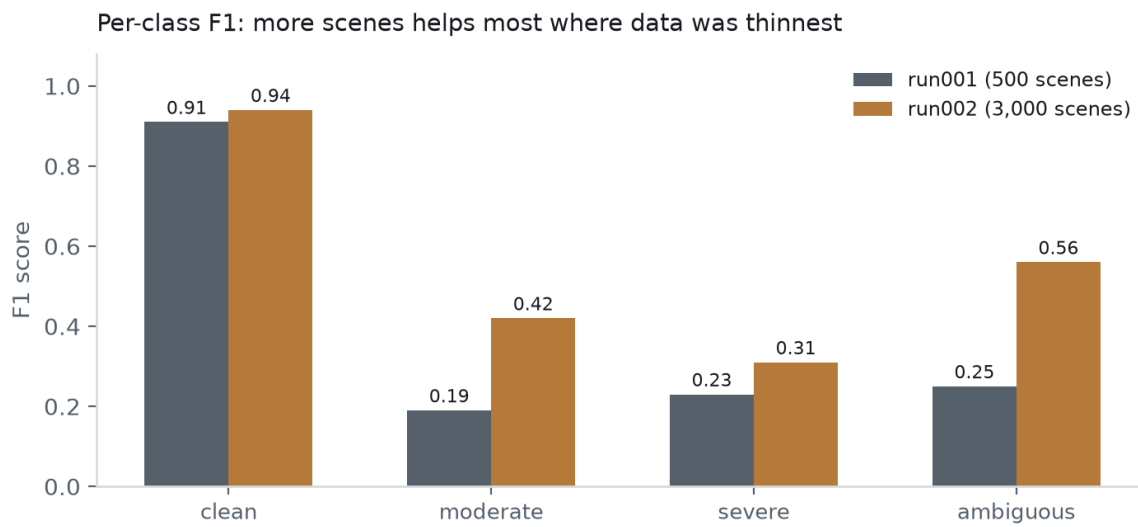


Figure 5: Per-class F1: run001 vs run002

is spread across both neighbors (moderate and clean) rather than concentrated at one boundary. Two hypotheses worth checking before assuming more data alone will fix it:

- The severe/moderate and severe/clean thresholds in `labeling.py` (`moderate_max: 0.15`, `severe_max: 0.5`) may not align well with what's visually/photometrically distinguishable — worth another calibration pass against thumbnails specifically in this range.
- Severe blends by definition span a wide continuous range of  $B$ ; the class may simply be intrinsically harder to separate from its neighbors than ambiguous is from clean, independent of sample count.

**Concrete next steps:**

- Scale further (10,000+ scenes) — cheap to test whether severe keeps improving or plateaus, which would distinguish the two hypotheses above.
- Revisit the severe-class thresholds with a targeted calibration pass.
- Bring in the 43 real LSST+Euclid cutouts as the actual sim-to-real check — still not done as of this report.